

[Appendix] SLeDGe: Semi-Supervised Learning on Data Streams via Graph Structure Learning

A Implementation Details

A.1 Experimental Environment

We conduct all experiments with RTX A6000 GPUs (48GB VRAM), 512GB of RAM, and Intel Xeon Silver 4210R Processors.

A.2 General Setup

We assume no initial labeled training set. The total memory size is fixed at 1,000 samples. Under the 1.0% label ratio configuration, we allocate $|L| = 100$ for labeled samples. For the 0.1% ratio, we define the labeled memory size as $|L| = \max(c, 10)$ to ensure at least one labeled sample per class, where c denotes the number of classes. In both scenarios, the remaining memory ($|U| = 1000 - |L|$) is reserved for unlabeled data.

For all deep learning models, we perform a hyperparameter grid search over the learning rate (lr), weight decay (wd), and time-decaying rate (τ). The search space for the learning rate spans $\{1e^{-5}, 5e^{-5}, \dots, 1e^{-2}, 5e^{-2}\}$, while weight decay is selected from $\{0.0, 5 \times 10^{-4}\}$. The time-decaying rate τ is chosen from $\{1, \infty\}$, where $\tau = \infty$ denotes the absence of temporal weighting, assigning uniform importance to every sample regardless of its timestamp

A.3 Baselines (Section 5.1 of the main paper)

- **TLP** is a graph-based semi-supervised learning method. We adopt the labeled and unlabeled sample counts as described above and follow the algorithm specified in the original paper.
- **ReSSL** is a cluster-based semi-supervised learning method. We use the authors' official implementation with its default hyperparameters unless otherwise specified. The memory size is set as described in our general setup.
- **HDSSL** is also a cluster-based semi-supervised learning method. Since we do not have an initial labeled training set, the Denoising Autoencoder (DAE) component could not be trained; therefore, we use only its clustering component. We set "chunkSize" = 1 and vary the number of neighbors $k \in \{1, 5, 10\}$. All other hyperparameters follow the original implementation.

A.4 Model architectures and ablations (Section 5.2 - Q2. Ablation Study of the main paper)

- **Backbone (*Base*)**: Our backbone model is an α_2 -layer MLP equipped with layer normalization ($\alpha_2 \in \{1, 2, 3\}$), chosen for its flexibility. This *Base* model operates without any memory modules. Labeled samples update the model via single-instance training, while unlabeled samples are used only for evaluation.

Table 9: Optimal hyperparameter configurations for SLeDGe.

Dataset	$\alpha_1 \in \{1, 2, 3\}$	$\alpha_2 \in \{1, 2, 3\}$	$k \in \{1, 5, 10\}$	$\tau \in \{1, \infty\}$
MNDS	2,1	2,1	5,10	1,1
Shopper	3,1	3,3	1,5	∞, ∞
WebKB	3,2	3,3	10,5	1,1
MNIST	3,3	2,1	5,5	∞, ∞
CIFAR-10	1,1	1,2	10,10	∞, ∞
KMNIST	3,3	1,3	5,5	∞, ∞
F.MNIST	1,1	3,2	10,5	∞, ∞
Shuttle	1,1	1,1	5,5	∞, ∞
GSAD	2,2	1,1	1,1	∞, ∞
SDD	2,2	2,1	1,1	1,1
HAR	2,2	1,1	10,10	∞, ∞
OD	1,2	1,1	1,1	1,1

- **+M**: This variant extends the *Base* model by introducing memory retention. When updating the memory buffers in this configuration, we compute cosine similarity using the raw input features rather than the learned embeddings (as detailed in Section 4.1).
- **Static**: Building upon the **+M** variant, this configuration incorporates a static k -nearest neighbors (k NN) graph for information propagation. This graph is constructed by calculating cosine similarity across the static input features. We search for the optimal number of neighbors $k \in \{1, 5, 10\}$.
- **+GSL**: To incorporate graph structure learning (GSL), we use an α_1 -layer MLP as the embedding function, where $\alpha_1 \in \{1, 2, 3\}$. In this ablation, the dynamic GSL operates without a regularization term. The number of neighbors k is selected from $\{1, 5, 10\}$.
- **SLeDGe**: Our proposed model utilizes the same GSL embedding function as above (an α_1 -layer MLP, $\alpha_1 \in \{1, 2, 3\}$). Unlike the **+GSL** variant, it includes a regularization term for dynamic GSL. The balancing parameter λ is selected from $\{0.001, 0.01, 0.1, 1.0\}$ and the number of $k \in \{1, 5, 10\}$.

A.5 Hyperparameter Configurations

We detail the specific hyperparameter settings used for SLeDGe across various datasets. SLeDGe relies on architectural parameters α_1 (GSL embedding layers), α_2 (Backbone MLP layers), the number of neighbors k , and the time-decaying rate τ . Unless otherwise specified, all experiments use the default configurations outlined in Table 9, where paired values denote the optimal settings for label ratios of 0.001 and 0.01, respectively. For other hyperparameters omitted from this table (e.g., lr, wd, λ), we conduct a separate search for each experiment.

Table 10: Average accuracy over the entire data stream. The best results are in **bold**, and the second best are underlined. SLeDGe achieves the highest accuracy for most datasets and label ratios.

Label	0.1%					1.0%				
Dataset	TLP	ReSSL	HDSSL	SLeDGe	SLeDGe-L	TLP	ReSSL	HDSSL	SLeDGe	SLeDGe-L
MNDS	0.0559 (0.0000)	0.0809 (0.0000)	0.0796 (0.0000)	<u>0.2792</u> (0.0138)	0.2974 (0.0225)	0.1182 (0.0126)	0.1915 (0.0300)	0.3199 (0.0250)	0.4415 (0.0300)	<u>0.3929</u> (0.0133)
Shopper	0.8453 (0.0000)	0.6304 (0.1475)	0.7571 (0.0554)	<u>0.8047</u> (0.0242)	0.7510 (0.0791)	0.8453 (0.0000)	0.7983 (0.0317)	0.8025 (0.0138)	<u>0.8437</u> (0.0025)	0.6768 (0.0232)
WebKB	0.1980 (0.0000)	0.2310 (0.0246)	0.9857 (0.0064)	<u>0.9943</u> (0.0005)	0.9946 (0.0005)	0.4005 (0.0580)	0.3223 (0.0368)	0.7179 (0.0312)	<u>0.9909</u> (0.0043)	0.9967 (0.0005)
MNIST	0.1773 (0.0076)	0.3400 (0.0585)	0.3797 (0.0765)	0.7086 (0.0109)	<u>0.6752</u> (0.0042)	0.1309 (0.0015)	0.6880 (0.0538)	0.7842 (0.0080)	0.8695 (0.0049)	<u>0.8102</u> (0.0094)
CIFAR-10	0.1123 (0.0034)	0.1064 (0.0122)	0.1173 (0.0081)	0.1519 (0.0092)	<u>0.1486</u> (0.0014)	0.1635 (0.0064)	0.1503 (0.0155)	0.1351 (0.0057)	0.2324 (0.0057)	<u>0.2126</u> (0.0062)
KMNIST	0.2290 (0.0176)	0.2293 (0.0493)	0.2768 (0.0506)	0.5498 (0.0148)	<u>0.5190</u> (0.0189)	0.4473 (0.0093)	0.5155 (0.0575)	0.6464 (0.0372)	0.7829 (0.0073)	<u>0.6781</u> (0.0066)
F.MNIST	0.1652 (0.0145)	0.3039 (0.0286)	0.3918 (0.0264)	<u>0.5286</u> (0.0277)	0.5700 (0.0220)	0.1458 (0.0052)	0.5809 (0.0219)	0.6596 (0.0088)	0.7126 (0.0103)	<u>0.6883</u> (0.0046)
Shuttle	0.7874 (0.0013)	0.8350 (0.0492)	0.8662 (0.0140)	0.6914 (0.0209)	0.7787 (0.0272)	0.7860 (0.0001)	<u>0.9640</u> (0.0030)	0.9672 (0.0047)	0.9370 (0.0132)	0.8519 (0.0360)
GSAD	0.2097 (0.0192)	0.1971 (0.0431)	0.2279 (0.0215)	0.3237 (0.0502)	<u>0.3031</u> (0.0534)	0.1933 (0.0208)	0.6267 (0.0306)	0.6718 (0.0267)	0.7553 (0.0121)	<u>0.6954</u> (0.0230)
SDD	0.7383 (0.0126)	0.6827 (0.0592)	0.5661 (0.0323)	<u>0.9967</u> (0.0001)	0.9984 (0.0001)	0.5497 (0.0444)	0.9587 (0.0094)	0.8239 (0.0097)	<u>0.9951</u> (0.0001)	0.9957 (0.0025)
HAR	0.2236 (0.0250)	0.3844 (0.0412)	0.2087 (0.0488)	0.6083 (0.0315)	<u>0.6057</u> (0.0664)	0.1988 (0.0051)	0.6032 (0.0387)	0.6682 (0.0472)	0.8242 (0.0090)	<u>0.7778</u> (0.0141)
OD	0.7683 (0.0119)	0.7717 (0.0274)	0.7801 (0.0127)	<u>0.7894</u> (0.0449)	0.8136 (0.0179)	0.7728 (0.0028)	0.8680 (0.0146)	<u>0.8841</u> (0.0063)	0.8951 (0.0167)	0.8673 (0.0080)
Average	0.3759	0.3892	0.4695	<u>0.6189</u>	0.6213	0.3960	0.6056	0.6734	0.7734	<u>0.7203</u>
Avg. Rank	4.00	4.00	3.25	1.83	<u>1.92</u>	4.42	3.75	3.00	1.42	<u>2.42</u>

B Additional Experiment Results

B.1 Performance over the Final Portions of the Data Stream (Section 5.2 - Q1. Performance of the main paper)

Table 10 summarizes the average accuracy and the standard deviation over 5 runs for all methods across datasets and label ratios. Furthermore, we evaluate model performance over the final 20%, 10%, and 5% portions of the data stream. The corresponding results are presented in Table 11, 12 and 13, respectively. Overall, SLeDGe maintains superior performance across datasets and label ratios, demonstrating its ability to effectively propagate supervision and adapt to evolving data distributions throughout the stream.

B.2 Memory Update Coefficient Analysis (Section 5.2 - Q2. Ablation Study of the main paper)

We investigate the effect of the memory update coefficients, α and β , on model performance. To validate the effectiveness of our adaptive update mechanism, we

Table 11: Average accuracy over the last 20% of the data stream. The best results are in **bold**, and the second best are underlined. SLeDGe achieves the highest accuracy for most datasets and label ratios.

Label	0.1%					1.0%				
Dataset	TLP	ReSSL	HDSSL	SLeDGe	SLeDGe-L	TLP	ReSSL	HDSSL	SLeDGe	SLeDGe-L
MNDS	0.0179	0.0000	0.0000	0.2819	<u>0.2297</u>	0.0744	0.1289	0.2151	<u>0.3727</u>	0.3997
Shopper	0.7997	0.6682	0.6707	0.7997	0.6550	0.7997	0.7622	0.7579	0.7997	0.6510
WebKB	0.0000	0.0302	0.7934	<u>0.9925</u>	0.9948	0.3650	0.2001	0.4124	<u>0.9804</u>	0.9960
MNIST	0.1648	0.4339	0.5102	0.7952	<u>0.7523</u>	0.1248	0.7826	<u>0.8964</u>	0.9177	0.8883
CIFAR-10	0.1144	0.1137	0.1183	<u>0.1623</u>	0.1656	0.1645	0.1704	0.1210	0.2595	<u>0.2398</u>
KMNIST	0.1895	0.2642	0.2668	0.5182	<u>0.4940</u>	0.4072	0.5413	0.7526	<u>0.7525</u>	0.6651
F.MNIST	0.1487	0.3853	0.5088	<u>0.5650</u>	0.6144	0.1484	0.6327	<u>0.7252</u>	0.7485	0.7128
Shuttle	0.7926	<u>0.9176</u>	0.9374	0.7485	0.8374	0.7915	<u>0.9798</u>	0.9868	0.9392	0.8811
GSAD	0.1813	0.1795	0.1557	0.3837	<u>0.3061</u>	0.1311	0.4825	0.4967	0.6434	<u>0.6247</u>
SDD	0.7797	0.7307	0.6639	<u>0.9965</u>	0.9981	0.9158	0.9761	0.8356	<u>0.9929</u>	0.9960
HAR	0.1876	0.4048	0.1769	0.6265	<u>0.6035</u>	0.1843	0.6646	0.7677	0.8918	<u>0.8329</u>
OD	0.7123	0.7404	<u>0.7555</u>	0.7904	0.7483	0.7332	0.8510	0.8488	0.8725	<u>0.8618</u>
Average	0.3407	0.4057	0.4631	0.6384	<u>0.6166</u>	0.4033	0.5977	0.6514	0.7642	<u>0.7291</u>
Avg. Rank	3.92	3.83	3.33	1.67	<u>2.08</u>	4.42	3.58	3.00	1.50	<u>2.42</u>

Table 12: Average accuracy over the last 10% of the data stream. The best results are in **bold**, and the second best are underlined. SLeDGe achieves the highest accuracy for most datasets and label ratios.

Label	0.1%					1.0%				
Dataset	TLP	ReSSL	HDSSL	SLeDGe	SLeDGe-L	TLP	ReSSL	HDSSL	SLeDGe	SLeDGe-L
MNDS	0.0110	0.0000	0.0000	0.2946	<u>0.2814</u>	0.0744	0.1923	0.2605	<u>0.3786</u>	0.3905
Shopper	0.7989	0.6567	0.6448	0.7989	0.6404	0.7989	0.7659	0.7494	0.7989	0.6397
WebKB	0.0000	0.0043	0.7773	<u>0.9906</u>	0.9937	0.3324	0.0507	0.1778	<u>0.9768</u>	0.9969
MNIST	0.1583	0.4364	0.5358	0.8146	<u>0.7659</u>	0.1265	0.7948	<u>0.9065</u>	0.9284	0.8967
CIFAR-10	0.1144	0.1159	0.1167	<u>0.1677</u>	0.1731	0.1581	0.1696	0.1532	0.2583	<u>0.2343</u>
KMNIST	0.1768	0.2543	0.2458	0.4720	<u>0.4530</u>	0.3984	0.5219	0.7343	<u>0.7215</u>	0.6250
F.MNIST	0.1424	0.3879	0.5124	<u>0.5710</u>	0.6144	0.1520	0.6308	<u>0.7243</u>	0.7473	0.7102
Shuttle	0.7995	<u>0.9180</u>	0.9406	0.7546	0.8414	0.7973	<u>0.9825</u>	0.9887	0.9399	0.8880
GSAD	0.2495	0.1686	0.2289	0.3965	<u>0.2960</u>	0.1983	0.4659	0.4348	0.6407	<u>0.6045</u>
SDD	0.9137	0.9384	0.8736	<u>0.9965</u>	0.9981	0.9998	0.9922	0.9661	0.9925	<u>0.9986</u>
HAR	0.1784	0.3934	0.1845	0.6150	<u>0.6058</u>	0.1732	0.6501	0.7701	0.8896	<u>0.8171</u>
OD	0.6926	0.7383	0.7328	0.8294	<u>0.7675</u>	0.7344	0.8980	0.8892	0.8811	<u>0.8903</u>
Average	0.3530	0.4177	0.4828	0.6418	<u>0.6192</u>	0.4120	0.5929	0.6462	0.7628	<u>0.7243</u>
Avg. Rank	4.17	3.50	3.50	1.67	<u>2.00</u>	4.08	3.42	3.08	1.83	<u>2.50</u>

compare the proposed dynamically computed coefficients against fixed values. As shown in Table 14, applying a high coefficient (e.g., 0.95) generally yields better performance compared to a low coefficient (e.g., 0.05), demonstrating superiority in 10 out of 12 datasets. However, unconditionally employing a high coefficient often leads to suboptimal results. For instance, on the CIFAR-10 dataset, our

Table 13: Average accuracy over the last 5% of the data stream. The best results are in **bold**, and the second best are underlined. SLeDGe achieves the highest accuracy for most datasets and label ratios.

Label	0.1%					1.0%				
Dataset	TLP	ReSSL	HDSSL	SLeDGe	SLeDGe-L	TLP	ReSSL	HDSSL	SLeDGe	SLeDGe-L
MNDS	0.0055	0.0000	0.0000	0.3949	<u>0.3336</u>	0.0976	0.1872	0.2804	<u>0.4125</u>	0.4209
Shopper	0.8214	0.6711	0.6601	0.8214	0.6455	0.8214	0.7841	0.7591	0.8214	0.6662
WebKB	0.0000	0.0087	0.7546	<u>0.9812</u>	0.9874	0.2937	0.0696	0.1952	<u>0.9715</u>	0.9937
MNIST	0.1600	0.4222	0.5516	0.8356	<u>0.7891</u>	0.1274	0.8054	<u>0.9274</u>	0.9502	0.9180
CIFAR-10	0.1121	0.1120	0.1101	<u>0.1702</u>	0.1750	0.1562	0.1695	0.1551	0.2549	<u>0.2315</u>
KMNIST	0.1740	0.2554	0.2495	0.4711	<u>0.4566</u>	0.4000	0.5359	0.7402	<u>0.7243</u>	0.6430
F.MNIST	0.1462	0.3901	0.5162	<u>0.5745</u>	0.6194	0.1498	0.6337	<u>0.7261</u>	0.7522	0.7127
Shuttle	0.7997	<u>0.9239</u>	0.9422	0.7659	0.8515	0.7953	<u>0.9862</u>	0.9898	0.9441	0.8950
GSAD	<u>0.3281</u>	0.1528	0.2414	0.4063	0.3258	0.1727	0.5951	0.4561	0.7447	<u>0.6521</u>
SDD	0.9154	1.0000	0.9488	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
HAR	0.1611	0.3615	0.1911	<u>0.5782</u>	0.5887	0.1455	0.6844	0.7899	0.9187	<u>0.8518</u>
OD	0.7967	0.8156	0.8189	0.8994	<u>0.8716</u>	0.8804	0.9198	0.9649	<u>0.9212</u>	0.9068
Average	0.3684	0.4261	0.4987	0.6582	0.6370	0.4200	0.6142	0.6654	0.7846	<u>0.7410</u>
Avg. Rank	4.00	3.42	3.50	1.67	<u>2.00</u>	4.08	3.33	<u>2.58</u>	1.50	<u>2.58</u>

Table 14: Effects of fixed versus dynamically computed memory update coefficients (α and β) on the accuracy of SLeDGe when label ratio is 1.0%. The proposed adaptive approach successfully ensures optimal accuracy, often outperforming the solid performance of a high fixed value.

Coefficient	α, β		
Dataset	0.05	0.95	Proposed
MNDS	0.3905	0.4080	0.3907
Shopper	0.8392	0.8385	0.8391
WebKB	0.9593	0.9628	0.9641
MNIST	0.7270	0.8332	0.7758
CIFAR-10	0.2315	0.2339	0.2471
KMNIST	0.5461	0.6412	0.6132
F.MNIST	0.6871	0.7256	0.7114
Shuttle	0.9415	0.9372	0.9403
GSAD	0.7978	0.8222	0.8104
SDD	0.9931	0.9957	0.9948
HAR	0.7939	0.7939	0.7960
OD	0.8885	0.9022	0.8994

proposed method achieves 0.2471, noticeably outperforming the 0.2339 obtained using a fixed high coefficient. These results confirm that our update mechanism is appropriately adaptive, offering a more reasonable approach than unconditionally using a high fixed value.

Table 15: SLeDGe results across varying labeled (L) and unlabeled (U) memory sizes. Increasing either memory size generally leads to performance gains.

Memory	L			U		
Dataset	20	100	Gain	180	900	Gain
MNDS	0.3904	0.4415	+ 13.09%	0.3816	0.4415	+ 15.70%
Shopper	0.8405	0.8437	+ 0.38%	0.8434	0.8437	+ 0.04%
WebKB	0.9922	0.9909	- 0.13%	0.9925	0.9909	- 0.16%
MNIST	0.8480	0.8695	+ 2.54%	0.8291	0.8695	+ 4.87%
CIFAR-10	0.2163	0.2324	+ 7.44%	0.2342	0.2324	- 0.77%
KMNIST	0.7290	0.7829	+ 7.39%	0.7325	0.7829	+ 6.88%
F.MNIST	0.6770	0.7126	+ 5.26%	0.7057	0.7126	+ 0.98%
Shuttle	0.7707	0.9370	+ 21.58%	0.9439	0.9370	- 0.73%
GSAD	0.7289	0.7553	+ 3.62%	0.7610	0.7553	- 0.75%
SDD	0.9946	0.9951	+ 0.05%	0.9944	0.9951	+ 0.07%
HAR	0.8140	0.8242	+ 1.25%	0.8051	0.8242	+ 2.37%
OD	0.8913	0.8951	+ 0.43%	0.8914	0.8951	+ 0.42%

B.3 Memory Size Analysis (Section 5.2 - Q4. Sensitivity of the main paper)

We evaluate the model’s performance by independently varying the labeled memory size (L) and the unlabeled memory size (U). To isolate the specific impact of each component, we perform a sensitivity analysis by varying only one size (L or U) at a time, keeping all other hyperparameters constant. Our baseline configuration uses $L = 100$ and $U = 900$ (reflecting the 1.0% label ratio setting). We test variations by reducing these values to 20% of their original levels (e.g., $L = 20$ or $U = 180$).

Table 15 shows that performance generally improves as memory size is expanded. Specifically, increasing L leads to performance enhancements across 11 out of 12 datasets, achieving an average relative gain of 5.24%. In contrast, increasing U yields improvements in 8 out of 12 datasets, resulting in a more modest average gain of 2.41%. These results indicate that SLeDGe exhibits higher sensitivity to the availability of labeled samples than to unlabeled data.